

大模型模块 - pytorch 手撕

1. 按 QKV 来源划分的 Attention

1.1 Self-Attention

自注意力机制。

```
import math
import torch
import torch.nn as nn
import torch.nn.functional as F

class SelfAttention(nn.Module):
    def __init__(self, d_model):
        super().__init__()
        self.d_model = d_model

        self.q_proj = nn.Linear(d_model, d_model, bias=False)
        self.k_proj = nn.Linear(d_model, d_model, bias=False)
        self.v_proj = nn.Linear(d_model, d_model, bias=False)
        self.o_proj = nn.Linear(d_model, d_model, bias=False)

    def forward(self, x, mask=None):
        """
        x: (batch, seq_len, d_model)
        mask: (seq_len, seq_len), True=屏蔽, broadcast到batch维度
        """
        # 计算QKV
        Q = self.q_proj(x) # Q: (batch, seq_len, d_model)
        K = self.k_proj(x) # K: (batch, seq_len, d_model)
        V = self.v_proj(x) # V: (batch, seq_len, d_model)

        # 计算注意力得分并添加掩码, scores: (batch, seq_len, seq_len)
        scores = torch.matmul(Q, K.transpose(-2, -1)) / math.sqrt(self.d_model)
        if mask is not None:
            scores = scores.masked_fill(mask, float("-inf"))

        # 计算注意力权重和输出
        attn_weights = F.softmax(scores, dim=-1) # attn_weights: (batch, seq_len, seq_len)
        out = torch.matmul(attn_weights, V) # out: (batch, seq_len, d_model)
        return self.o_proj(out) # return: (batch, seq_len, d_model)
```

外部调用 demo:

```
import torch
batch = 2
seq_len = 4
d_model = 8
```

```
# 不传mask的版本
x = torch.randn(batch, seq_len, d_model)
attn = SelfAttention(d_model=d_model)
out = attn(x)

# 传mask的版本
x = torch.randn(batch, seq_len, d_model)
mask = torch.triu(torch.ones(seq_len, seq_len, dtype=torch.bool), diagonal=1)
out = attn(x, mask=mask)
```

pytorch 相关基础

- `super().__init__()` 调用所继承父类的初始化方法。
- 线性层 `self.W = nn.Linear(...)` 被 `self.W(x)` 调用时，实际会对 `x` 的最后一维执行 $x @ W^T + b$ 。
- `if mask is not None` 不能替换为 `if mask`，因为 `mask` 本身是布尔型张量，如果直接 `if mask`，pytorch 会直接报错：Boolean value of Tensor with more than one value is ambiguous。
- `torch.triu()` 建立上三角矩阵，`diagonal=1` 保留主对角线上方（不含主对角线），`diagonal=0` 保留主对角线及其上方（包含主对角线），`diagonal=-1` 保留主对角线下方1条线及其上方。这里对于某个当前 token 不应该对自己加掩码，因此应设置 `mask=False`。

1.2 Cross-Attention

交叉注意力机制。

```
import math
import torch
import torch.nn as nn
import torch.nn.functional as F

class CrossAttention(nn.Module):
    def __init__(self, d_model):
        super().__init__()
        self.d_model = d_model

        self.q_proj = nn.Linear(d_model, d_model, bias=False)
        self.k_proj = nn.Linear(d_model, d_model, bias=False)
        self.v_proj = nn.Linear(d_model, d_model, bias=False)
        self.o_proj = nn.Linear(d_model, d_model, bias=False)

    def forward(self, x_q, x_kv, mask=None):
        """
        x_q: (batch, seq_q, d_model)
        x_kv: (batch, seq_kv, d_model)
        mask: (seq_q, seq_kv), True=屏蔽, broadcast 到batch维度
        """
        # 计算QKV
        Q = self.q_proj(x_q) # Q: (batch, seq_q, d_model)
        K = self.k_proj(x_kv) # K: (batch, seq_kv, d_model)
        V = self.v_proj(x_kv) # V: (batch, seq_kv, d_model)

        # 计算注意力得分并添加掩码, scores: (batch, seq_q, seq_kv)
```

```

scores = torch.matmul(Q, K.transpose(-2, -1)) / math.sqrt(self.d_model)
if mask is not None:
    scores = scores.masked_fill(mask, float("-inf"))

# 计算注意力权重和输出
attn_weights = F.softmax(scores, dim=-1) # attn_weights: (batch, seq_q, seq_kv)
out = torch.matmul(attn_weights, V) # out: (batch, seq_q, d_model)

return self.o_proj(out) # return: (batch, seq_len, d_model)

```

外部调用，如果添加 `mask` 一般是 `padding mask`：

```

batch = 2
seq_q = 3
seq_kv = 5
d_model = 8

x_q = torch.randn(batch, seq_q, d_model)
x_kv = torch.randn(batch, seq_kv, d_model)

cross_attn = CrossAttention(d_model)
out = cross_attn(x_q, x_kv)

```

2. 按 KV 组织方式划分的 Attention

2.1 Multi-Head Self-Attention (MHA)

多头自注意力机制。

```

import math
import torch
import torch.nn as nn
import torch.nn.functional as F

class MultiHeadAttention(nn.Module):
    def __init__(self, d_model, num_heads):
        super().__init__()
        assert d_model % num_heads == 0

        self.d_model = d_model
        self.num_heads = num_heads
        self.head_dim = d_model // num_heads

        self.q_proj = nn.Linear(d_model, d_model, bias=False)
        self.k_proj = nn.Linear(d_model, d_model, bias=False)
        self.v_proj = nn.Linear(d_model, d_model, bias=False)
        self.o_proj = nn.Linear(d_model, d_model, bias=False)

    def forward(self, x, mask=None):
        """
        x: (batch, seq_len, d_model)
        mask: (seq_len, seq_len), True=屏蔽, broadcast到batch和num_heads维度
        """

```

```

"""
# 获取x的相关维度
batch, seq_len, _ = x.shape

# 计算QKV
Q = self.q_proj(x) # Q:(batch, seq_len, d_model)
K = self.k_proj(x) # K:(batch, seq_len, d_model)
V = self.v_proj(x) # V:(batch, seq_len, d_model)

# QKV分头: (batch, num_heads, seq_len, head_dim)
Q = Q.view(batch, seq_len, self.num_heads, self.head_dim).transpose(1, 2)
K = K.view(batch, seq_len, self.num_heads, self.head_dim).transpose(1, 2)
V = V.view(batch, seq_len, self.num_heads, self.head_dim).transpose(1, 2)

# 计算注意力得分并添加掩码, scores:(batch, num_heads, seq_len, seq_len)
scores = torch.matmul(Q, K.transpose(-2, -1)) / math.sqrt(self.head_dim)
if mask is not None:
    scores = scores.masked_fill(mask, float('-inf'))

# 计算注意力权重和输出
attn_weights = F.softmax(scores, dim=-1) # attn_weights: (batch, num_heads,
seq_len, seq_len)
out = torch.matmul(attn_weights, V) # out: (batch, num_heads, seq_len, head_dim)

# 合并多头结果
out = out.transpose(1, 2) # out: (batch, seq_len, num_heads, head_dim)
out = out.contiguous().view(batch, seq_len, self.d_model) # out: (batch, seq_len,
d_model)

return self.o_proj(out) # return: (batch, seq_len, d_model)

```

外部调用:

```

x = torch.randn(2, 4, 8)
mha = MultiHeadAttention(d_model=8, num_heads=2)
mask = torch.triu(torch.ones(4, 4, dtype=torch.bool), diagonal=1)
out = mha(x, mask=mask)

```

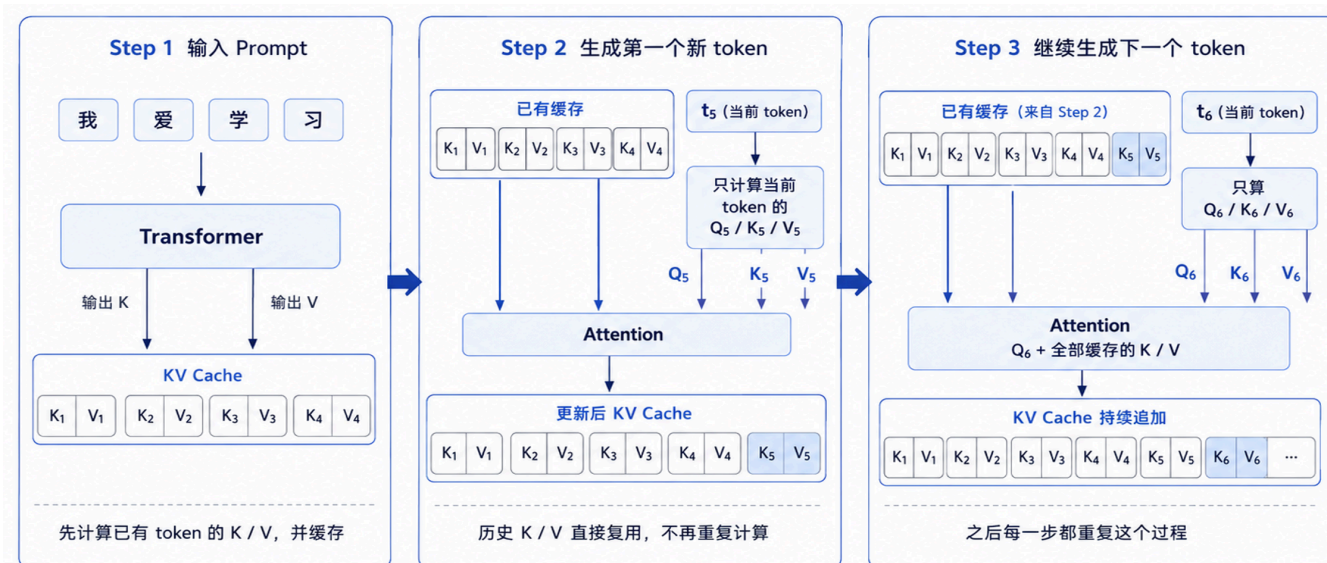
pytorch 相关基础:

- 如果 `assert` 希望给用户提示, 可以写 `assert d_model % num_heads == 0, "d_model must be a multiples of num_heads"`。
- 如果传入的 `mask` 是 `(batch, seq_len, seq_len)` 的形状 (通过 `if mask.dim() == 3` 来判断), 需要先执行 `mask.unsqueeze(1)` 变为 `(batch, 1, seq_len, seq_len)`, 否则广播机制对齐会出错。

2.2 MHA with KV Cache

带 KV Cache 的多头注意力机制。

使用 KV Cache 的自回归生成中, 每步只计算当前 token 的 Q/K/V, 历史 token 的 K/V 保存在缓存中复用, 从而避免重复计算加速推理。



```
import math
import torch
import torch.nn as nn
import torch.nn.functional as F

class MultiHeadAttention(nn.Module):
    def __init__(self, d_model, num_heads):
        super().__init__()
        assert d_model % num_heads == 0
        self.d_model = d_model
        self.num_heads = num_heads
        self.head_dim = d_model // num_heads

        self.q_proj = nn.Linear(d_model, d_model, bias=False)
        self.k_proj = nn.Linear(d_model, d_model, bias=False)
        self.v_proj = nn.Linear(d_model, d_model, bias=False)
        self.o_proj = nn.Linear(d_model, d_model, bias=False)

    def forward(self, x, mask=None, past_kv=None):
        """
        x: (batch, seq_len, d_model)
        mask: (seq_len, past_len+seq_len), True=屏蔽, broadcast到batch和num_heads维度
        past_kv: tuple, (past_k, past_v)
        past_k/past_v: (batch, num_heads, past_len, head_dim)
        """
        # 获取x相关维度
        batch, seq_len, _ = x.shape

        # 计算QKV
        Q = self.q_proj(x) # Q: (batch, seq_len, d_model)
        K = self.k_proj(x) # K: (batch, seq_len, d_model)
        V = self.v_proj(x) # V: (batch, seq_len, d_model)

        # QKV分头: (batch, num_heads, seq_len, head_dim)
        Q = Q.view(batch, seq_len, self.num_heads, self.head_dim).transpose(1, 2)
        K = K.view(batch, seq_len, self.num_heads, self.head_dim).transpose(1, 2)
        V = V.view(batch, seq_len, self.num_heads, self.head_dim).transpose(1, 2)
```

```

# 使用KV Cache
if past_kv is not None:
    past_k, past_v = past_kv
    # 新旧KV在seq_len维度拼接: (batch, num_heads, past_len+seq_len, head_dim)
    k = torch.cat([past_k, K], dim=-2)
    v = torch.cat([past_v, V], dim=-2)
new_kv = (K, V) # 更新KV Cache

# 计算注意力得分并添加掩码, scores:(batch, num_heads, seq_len, past_len+seq_len)
scores = torch.matmul(Q, K.transpose(-2, -1)) / math.sqrt(self.head_dim)
if mask is not None:
    scores = scores.masked_fill(mask, float('-inf'))

# 计算注意力权重和输出
attn_weights = F.softmax(scores, dim=-1) # attn_weights: (batch, num_heads,
seq_len, past_len+seq_len)
out = torch.matmul(attn_weights, V) # out: (batch, num_heads, seq_len, head_dim)

# 合并多头结果
out = out.transpose(1, 2) # out: (batch, seq_len, num_heads, head_dim)
out = out.contiguous().view(batch, seq_len, self.d_model) # out: (batch, seq_len,
d_model)

# return: (batch, seq_len, d_model), KV Cache
return self.o_proj(out), new_kv

```

外部调用:

```

batch = 2
d_model = 8
num_heads = 2
steps = 5

mha = MultiHeadAttention(d_model=d_model, num_heads=num_heads)
past_kv = None

for t in range(steps):
    # 自回归推理时, 每一步只输入当前新 token
    x = torch.randn(batch, 1, d_model)
    out, past_kv = mha(x, past_kv=past_kv)

    print(f"step {t + 1}")
    print("out:", out.shape)
    print("cache_k:", past_kv[0].shape)
    print("cache_v:", past_kv[1].shape)

```

注意循环中不断传入 `past_kv`, 结果输出大致如下:

```

step 1
out:      torch.Size([2, 1, 8])
cache_k: torch.Size([2, 2, 1, 4])

```

```
cache_v: torch.Size([2, 2, 1, 4])
```

step 2

```
out:      torch.Size([2, 1, 8])
cache_k: torch.Size([2, 2, 2, 4])
cache_v: torch.Size([2, 2, 2, 4])
```

step 3

```
out:      torch.Size([2, 1, 8])
cache_k: torch.Size([2, 2, 3, 4])
cache_v: torch.Size([2, 2, 3, 4])
```

从代码上，`past_kv` 保存了所有过去 token 的 KV。如果不使用 KV Cache，每次传入的 `x` 的 `seq_len` 会越来越长，因为每次 `x` 都得带上之前已经生成的 token 前缀，然后重新计算它们的 KV；使用 KV Cache 后，每次传入的 `x` 维度为 `(batch, 1, d_model)`。

2.3 Multi-Query Attention (MQA)

多查询注意力机制。Q 使用 `num_heads` 个头，KV 使用 1 个头。

```
import math
import torch
import torch.nn as nn
import torch.nn.functional as F

class MultiQueryAttention(nn.Module):
    def __init__(self, d_model, num_heads):
        super().__init__()
        assert d_model % num_heads == 0
        self.d_model = d_model
        self.num_heads = num_heads
        self.head_dim = d_model // num_heads

        # KV只有一个头的参数量
        self.q_proj = nn.Linear(d_model, d_model, bias=False)
        self.k_proj = nn.Linear(d_model, self.head_dim, bias=False)
        self.v_proj = nn.Linear(d_model, self.head_dim, bias=False)
        self.o_proj = nn.Linear(d_model, d_model, bias=False)

    def forward(self, x, mask=None):
        """
        x: (batch, seq_len, d_model)
        mask: (seq_len, seq_len), True=屏蔽, broadcast到batch和num_heads维度
        """
        # 获取x相关维度
        batch, seq_len, _ = x.shape

        # 计算QKV
        Q = self.q_proj(x) # Q: (batch, seq_len, d_model)
        K = self.k_proj(x) # K: (batch, seq_len, head_dim)
        V = self.v_proj(x) # V: (batch, seq_len, head_dim)
```

```

# QKV分头, Q: (batch, num_heads, seq_len, head_dim)
Q = Q.view(batch, seq_len, self.num_heads, self.head_dim).transpose(1, 2)
K = K.unsqueeze(1) # K: (batch, 1, seq_len, head_dim)
V = V.unsqueeze(1) # V: (batch, 1, seq_len, head_dim)

# 自动广播K计算注意力得分并添加掩码, scores:(batch, num_heads, seq_len, seq_len)
scores = torch.matmul(Q, K.transpose(-2, -1)) / math.sqrt(self.head_dim)
if mask is not None:
    scores = scores.masked_fill(mask, float("-inf")) # mask: (batch, 1, seq_len,
seq_len)

# 计算注意力权重和输出
attn_weights = F.softmax(scores, dim=-1) # attn_weights: (batch, num_heads,
seq_len, seq_len)
out = torch.matmul(attn_weights, V) # out: (batch, num_heads, seq_len, head_dim)

# 合并多头结果
out = out.transpose(1, 2) # out: (batch, seq_len, num_heads, head_dim)
out = out.contiguous().view(batch, seq_len, self.d_model) # out: (batch, seq_len,
d_model)

return self.o_proj(out) # return: (batch, seq_len, d_model)

```

外部调用:

```

batch = 2
seq_len = 4
d_model = 8
num_heads = 2

x = torch.randn(batch, seq_len, d_model)
mqa = MultiQueryAttention(d_model=d_model, num_heads=num_heads)
mask = torch.triu(torch.ones(seq_len, seq_len, dtype=torch.bool), diagonal=1)
out = mqa(x, mask=mask)

```

2.4 Grouped-Query Attention (GQA)

分组查询注意力机制。

```

import math
import torch
import torch.nn as nn
import torch.nn.functional as F

class GroupedQueryAttention(nn.Module):
    def __init__(self, d_model, num_q_heads, num_kv_heads):
        super().__init__()
        assert d_model % num_q_heads == 0
        assert num_q_heads % num_kv_heads == 0
        self.d_model = d_model
        self.num_q_heads = num_q_heads
        self.num_kv_heads = num_kv_heads

```



```

self.head_dim = d_model // num_q_heads
self.num_groups = num_q_heads // num_kv_heads

self.q_proj = nn.Linear(d_model, d_model, bias=False)
# kv的线性层映射到 head_dim * num_kv_heads 维
self.k_proj = nn.Linear(d_model, self.head_dim * self.num_kv_heads, bias=False)
self.v_proj = nn.Linear(d_model, self.head_dim * self.num_kv_heads, bias=False)
self.o_proj = nn.Linear(d_model, d_model, bias=False)

def forward(self, x, mask=None):
    """
    x: (batch, seq_len, d_model)
    mask: (seq_len, seq_len), True=屏蔽, broadcast到batch和num_q_heads维度
    """
    # 获取x相关维度
    batch, seq_len, _ = x.shape

    # 计算QKV
    Q = self.q_proj(x) # Q: (batch, seq_len, d_model)
    K = self.k_proj(x) # K: (batch, seq_len, head_dim*num_kv_heads)
    V = self.v_proj(x) # V: (batch, seq_len, head_dim*num_kv_heads)

    # Q分头, Q: (batch, num_q_heads, seq_len, head_dim)
    Q = Q.view(batch, seq_len, self.num_q_heads, self.head_dim).transpose(1, 2)
    # KV分头, KV: (batch, num_kv_heads, seq_len, head_dim)
    K = K.view(batch, seq_len, self.num_kv_heads, self.head_dim).transpose(1, 2)
    V = V.view(batch, seq_len, self.num_kv_heads, self.head_dim).transpose(1, 2)

    # 扩展KV: (batch, num_q_heads, seq_len, head_dim)
    K = K.repeat_interleave(self.num_groups, dim=1)
    V = V.repeat_interleave(self.num_groups, dim=1)

    # 计算注意力得分并添加掩码, scores: (batch, num_q_heads, seq_len, seq_len)
    scores = torch.matmul(Q, K.transpose(-2, -1)) / math.sqrt(self.head_dim)
    if mask is not None:
        scores = scores.masked_fill(mask, float('-inf'))

    # 计算注意力权重和输出
    attn_weights = F.softmax(scores, dim=-1) # attn_weights: (batch, num_q_heads,
seq_len, seq_len)
    out = torch.matmul(attn_weights, V) # out: (batch, num_q_heads, seq_len, head_dim)

    # 合并多头结果
    out = out.transpose(1, 2)
    out = out.contiguous().view(batch, seq_len, self.d_model)

    return self.o_proj(out) # return: (batch, seq_len, d_model)

```

外部调用:

```

batch = 2
seq_len = 4
d_model = 8

```

```

num_q_heads = 4
num_kv_heads = 2
x = torch.randn(batch, seq_len, d_model)

gqa = GroupedQueryAttention(
    d_model=d_model,
    num_q_heads=num_q_heads,
    num_kv_heads=num_kv_heads
)

mask = torch.triu(
    torch.ones(seq_len, seq_len, dtype=torch.bool),
    diagonal=1,
)

out = gqa(x, mask=mask)

```

pytorch 相关基础

- **K** 和 **V** 的 head 维度不是1，因此不能自动广播到 **Q** 的 head 维度。

2.5 Multi-head Latent Attention (MLA)

多头潜变量注意力机制，配合 KV Cache 使用，将 KV 压缩为潜变量存储，需要计算时再恢复原 KV 表示，从而降低显存占用。

```

import math
import torch
import torch.nn as nn
import torch.nn.functional as F

class MultiHeadLatentAttention(nn.Module):
    def __init__(self, d_model, num_heads, latent_dim):
        super().__init__()
        assert d_model % num_heads == 0
        self.d_model = d_model
        self.num_heads = num_heads
        self.head_dim = d_model // num_heads

        self.q_proj = nn.Linear(d_model, d_model, bias=False)
        self.kv_down_proj = nn.Linear(d_model, latent_dim, bias=False)
        self.k_up_proj = nn.Linear(latent_dim, d_model, bias=False)
        self.v_up_proj = nn.Linear(latent_dim, d_model, bias=False)
        self.o_proj = nn.Linear(d_model, d_model, bias=False)

    def forward(self, x, mask=None, past_latent_kv=None):
        """
        x: (batch, seq_len, d_model)
        mask: (seq_len, past_len + seq_len)
        past_latent_kv: None或(batch, past_len+seq_len, latent_dim)
        """

        # 获取x相关维度
        batch, seq_len, _ = x.shape

```

```

# 计算QKV
Q = self.q_proj(x) # Q: (batch, seq_len, d_model)
latent_kv = self.kv_down_proj(x) # K: (batch, seq_len, latent_dim)

# 使用KV Cache
past_len = past_latent_kv.size(-2) if past_latent_kv is not None else 0
if past_latent_kv is not None:
    # 新旧latent_kv在seq_len维度拼接: (batch, past_len+seq_len, latent_dim)
    latent_kv = torch.cat([past_latent_kv, latent_kv], dim=-2)
new_latent_kv = latent_kv

# 潜变量还原
K = self.k_up_proj(latent_kv) # K: (batch, past_len+seq_len, d_model)
V = self.v_up_proj(latent_kv) # V: (batch, past_len+seq_len, d_model)

# QKV分头, Q: (batch, num_heads, seq_len, head_dim)
Q = Q.view(batch, seq_len, self.num_heads, self.head_dim).transpose(1, 2)
# KV: (batch, num_heads, past_len + seq_len, head_dim)
K = K.view(batch, past_len + seq_len, self.num_heads, self.head_dim).transpose(1, 2)
V = V.view(batch, past_len + seq_len, self.num_heads, self.head_dim).transpose(1, 2)

# 计算注意力得分并添加掩码, scores: (batch, num_heads, seq_len, past_len + seq_len)
scores = torch.matmul(Q, K.transpose(-2, -1)) / math.sqrt(self.head_dim)
if mask is not None:
    scores = scores.masked_fill(mask, float('-inf'))

# 计算注意力权重和输出
attn_weights = F.softmax(scores, dim=-1) # attn_weights: (batch, num_heads,
seq_len, past_len + seq_len)
out = torch.matmul(attn_weights, V) # out: (batch, num_heads, seq_len, head_dim)

# 合并多头结果
out = out.transpose(1, 2) # out: (batch, seq_len, num_heads, head_dim)
out = out.contiguous().view(batch, seq_len, self.d_model)

# return: (batch, seq_len, d_model), (batch, past_len + seq_len, latent_dim)
return self.o_proj(out), new_latent_kv

```

3. Transformer

3.1 LayerNorm

层归一化。

```

import torch
import torch.nn as nn

class LayerNorm(nn.Module):
    def __init__(self, d_model, eps=1e-5):
        super().__init__()
        self.eps = eps

```

```

# 初始化gamma=1,beta=0
self.gamma = nn.Parameter(torch.ones(d_model))
self.beta = nn.Parameter(torch.zeros(d_model))

def forward(self, x):
    """
    x: (batch, seq_len, d_model)
    """
    # 计算均值和方差
    mean = x.mean(dim=-1, keepdim=True)
    var = x.var(dim=-1, keepdim=True, unbiased=False)

    # 归一化并缩放平移
    x = (x - mean) / torch.sqrt(var + self.eps)
    x = self.gamma * x + self.beta
    return x

```

pytorch 相关基础

- 对张量某一维特征计算统计量聚合时，`keepdim=True` 保证聚合后该维度为1，并不会消失而减少维度。

3.2 RMSNorm

Root Mean Square Layer Normalization，均方根归一化。

```

import torch
import torch.nn as nn

class RMSNorm(nn.Module):
    def __init__(self, d_model, eps=1e-5):
        super().__init__()
        self.eps = eps
        self.gamma = nn.Parameter(torch.ones(d_model))

    def forward(self, x):
        """
        x: (batch, seq_len, d_model)
        """
        # 计算均方根
        rms = torch.sqrt(torch.mean(x ** 2, dim=-1, keepdim=True) + self.eps)

        # 归一化并缩放
        x = x / rms
        x = self.gamma * x
        return x

```

3.3 RoPE

旋转位置编码。

对第 m 个位置、某一对维度 $(2i, 2i + 1)$ ，RoPE 做二维旋转：

$$\begin{pmatrix} x'_{2i} \\ x'_{2i+1} \end{pmatrix} = \begin{pmatrix} \cos(m\theta_i) & -\sin(m\theta_i) \\ \sin(m\theta_i) & \cos(m\theta_i) \end{pmatrix} \begin{pmatrix} x_{2i} \\ x_{2i+1} \end{pmatrix}$$

其中

$$\theta_i = 10000^{-2i/d}$$

展开就是：

$$\begin{aligned} x'_{2i} &= x_{2i} \cos(m\theta_i) - x_{2i+1} \sin(m\theta_i) \\ x'_{2i+1} &= x_{2i} \sin(m\theta_i) + x_{2i+1} \cos(m\theta_i) \end{aligned}$$

对 Q, K 都这样旋转。其关键性质是：

$$\langle R_m q, R_n k \rangle = \langle q, R_{n-m} k \rangle$$

所以注意力里自然编码了相对位置 $n - m$ 。

```
import torch
import torch.nn as nn

class RoPE(nn.Module):
    def __init__(self, d_model, base=10000):
        super().__init__()
        assert d_model % 2 == 0

        self.d_model = d_model
        self.base = base

    def forward(self, x):
        """
        (x: batch, seq_len, d_model)
        """
        # 获取x相关信息
        seq_len = x.shape[-2]
        device = x.device

        # 生成频率, freq: (d_model / 2,)
        dim = torch.arange(0, self.d_model, 2, device=device) # dim: (d_model / 2,)
        freq = self.base ** (-dim / self.d_model)

        # 生成旋转角, theta: (seq_len, d_model / 2)
        pos = torch.arange(seq_len, device=device) # pos: (seq_len,)
        theta = torch.outer(pos, freq)

        # 计算正弦余弦值, cos/sin: (seq_len, d_model / 2)
        cos = torch.cos(theta)
        sin = torch.sin(theta)

        # x拆分为奇偶维, x_odd/x_even: (batch, seq_len, d_model / 2)
        x_even = x[..., 0::2]
        x_odd = x[..., 1::2]
```

```

# 计算奇偶维RoPE, out_odd, out_even: (batch, seq_len, d_model / 2)
out_even = cos * x_even - sin * x_odd
out_odd = sin * x_even + cos * x_odd

# 合并奇偶维, out: (batch, seq_len, d_model)
out = torch.zeros_like(x)
out[..., 0::2] = out_even
out[..., 1::2] = out_odd
return out

```

- 使用 `torch.arange`, `torch.randn`, `torch.ones`, `torch.zeros` 等创建新张量（例如本题中的 `dim` 和 `pos`）时，如果不指定 `device` 默认创建在 `cpu`，而一般传入的张量 `x` 会在 `gpu` 上，如果不指定 `device=device`，把新张量放在跟 `x` 同样的设备上，就会报错：Expected all tensors to be on the same device。如果使用 `torch.zeros_like(x)`，则会直接继承 `x` 的 `shape`, `device`, `dtype`, `layout`。
- `...` 可以省略部分维度，可以出现在开头、中间、结尾，但一个索引表达式里只能有一个。

自注意力机制调用 RoPE：

```

import math
import torch
import torch.nn as nn
import torch.nn.functional as F

class RoPE(nn.Module):
    """
    使用上面定义好的RoPE类，此处省略
    """

class SelfAttention(nn.Module):
    def __init__(self, d_model):
        super().__init__()
        assert d_model % 2 == 0

        self.d_model = d_model

        self.q_proj = nn.Linear(d_model, d_model, bias=False)
        self.k_proj = nn.Linear(d_model, d_model, bias=False)
        self.v_proj = nn.Linear(d_model, d_model, bias=False)
        self.o_proj = nn.Linear(d_model, d_model, bias=False)

        self.rope = RoPE(d_model)

    def forward(self, x, mask=None):
        """
        x: (batch, seq_len, d_model)
        mask: (seq_len, seq_len), True=屏蔽, broadcast到batch维度
        """
        Q = self.q_proj(x)
        K = self.k_proj(x)
        V = self.v_proj(x)

```

```

# RoPE 只作用在 Q 和 K 上
Q = self.rope(Q)
K = self.rope(K)

scores = torch.matmul(Q, K.transpose(-2, -1)) / math.sqrt(self.d_model)
if mask is not None:
    scores = scores.masked_fill(mask, float("-inf"))

attn_weights = F.softmax(scores, dim=-1)
out = torch.matmul(attn_weights, V)

return self.o_proj(out)

```

3.4 SwiGLU

$$\text{SwiGLU}(x) = \text{SiLU}(xW_1) \odot (xW_3)$$

再接一个输出投影形成 FFN:

$$\text{FFN}(x) = [\text{SiLU}(xW_1) \odot (xW_3)] W_2$$

其中:

- xW_1 : 门控分支
- xW_2 : 值分支
- $\odot$: 逐元素乘法
- $\text{SiLU}(z) = z\sigma(z)$

```

import torch
import torch.nn as nn
import torch.nn.functional as F

class SwiGLU(nn.Module):
    def __init__(self, d_model, d_ff):
        super().__init__()
        self.w1 = nn.Linear(d_model, d_ff, bias=False) # gate_proj
        self.w3 = nn.Linear(d_model, d_ff, bias=False) # up_proj
        self.w2 = nn.Linear(d_ff, d_model, bias=False) # down_proj

    def forward(self, x):
        """
        x: (batch, seq_len, d_model)
        """
        # 计算gate分支和up分支
        gate = F.silu(self.w1(x)) # gate: (batch, seq_len, d_ff)
        value = self.w3(x) # value: (batch, seq_len, d_ff)

        out = gate * value # 逐元素相乘
        out = self.w2(out)
        return out

```

3.5 Transformer

定义 Transformer 单个模块以及整个 Decoder-only Transformer 架构。Transformer 原文架构如下。

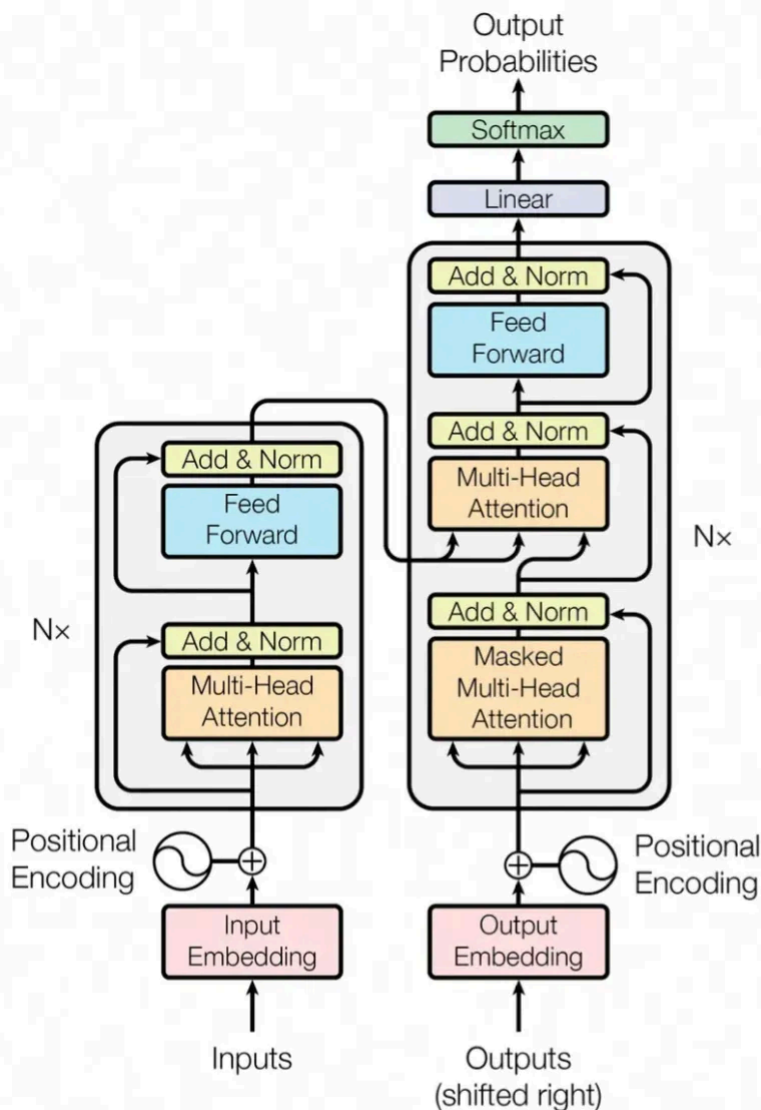


Figure 1: The Transformer - model architecture.

原文使用的是 Post-LN，下面代码实现使用现在更常用的 Pre-LN，梯度更稳定。另外，位置编码这里使用的是可学习编码，如果使用 RoPE，参考 3.3 小节直接修改调用的多头注意力类即可，不需要改动下面的代码（当然可学习编码要去掉）。

```
import torch
import torch.nn as nn
# 自定义模块部分
from attention import MultiHeadAttention
from norm import RMSNorm
from ffn import SwiGLU

class TransformerBlock(nn.Module):
    def __init__(self, d_model, num_heads, d_ff):
        super().__init__()

        self.attn = MultiHeadAttention(d_model, num_heads)
```



```

self.ffn = SwiGLU(d_model, d_ff)

self.norm1 = RMSNorm(d_model)
self.norm2 = RMSNorm(d_model)

def forward(self, x, mask=None):
    """
    x: (batch, seq_len, d_model)
    mask: (seq_len, seq_len), causal mask / padding mask
    """
    # Pre-LN + MHA + Residual Connection
    x = x + self.attn(self.norm1(x), mask=mask)
    # Pre-LN + FFN + Residual Connection
    x = x + self.ffn(self.norm2(x))
    return x

class DecoderOnlyTransformer(nn.Module):
    def __init__(self, vocab_size, d_model, num_heads, d_ff, num_layers, max_seq_len):
        super().__init__()

        self.token_emb = nn.Embedding(vocab_size, d_model) # 词嵌入
        self.pos_emb = nn.Embedding(max_seq_len, d_model) # 可学习的位置编码

        self.layers = nn.ModuleList(
            [TransformerBlock(d_model, num_heads, d_ff) for _ in range(num_layers)]
        ) # 注意力层

        self.norm = RMSNorm(d_model) # 归一化层
        self.lm_head = nn.Linear(d_model, vocab_size, bias=False) # 输出头

    def forward(self, input_ids, mask=None):
        """
        input_ids: (batch, seq_len)
        mask: (seq_len, seq_len)
        """
        # 获取input_ids相关信息
        seq_len = input_ids.shape[-1]
        device = input_ids.device

        # 词嵌入并添加可学习位置编码
        pos = torch.arange(seq_len, device=device) # pos: (seq_len,)
        pos = pos.unsqueeze(0) # pos: (1, seq_len), broadcast到batch维度
        x = self.token_emb(input_ids) + self.pos_emb(pos) # x: (batch, seq_len, d_model)

        # 注意力层处理, 处理过程中始终保持x: (batch, seq_len, d_model)
        for layer in self.layers:
            x = layer(x, mask=mask)

        x = self.norm(x) # 归一化层
        logits = self.lm_head(x) # 输出头, logits: (batch, seq_len, vocab_size)

        return logits

```

定义 `top-k` 采样方法（在后续解码策略部分会再次提到该方法），之后进行外部调用：

```
import torch
import torch.nn.functional as F

def top_k_sampling(logits, k=50, temperature=1.0):
    """
    logits: (batch, vocab_size)
    return: (batch, 1)
    """
    if temperature <= 0: # temperature=0即贪心搜索
        return torch.argmax(logits, dim=-1, keepdim=True)

    # 温度+top-k采样, topk_logits/idx: (batch, k)
    logits = logits / temperature
    topk_logits, topk_idx = torch.topk(logits, k, dim=-1)

    # 只对top-k进行softmax归一化，并取下标映射回原始id
    probs = F.softmax(topk_logits, dim=-1) # probs: (batch, k)
    sampled_idx = torch.multinomial(probs, num_samples=1) # sampled_idx: (batch, 1)
    next_token = torch.gather(topk_idx, dim=-1, index=sampled_idx) # next_token: (batch, 1)
    return next_token
```

外部调用：

```
batch = 2
seq_len = 8
vocab_size = 10000
d_model = 512
num_heads = 8
d_ff = 2048
num_layers = 6
max_seq_len = 1024

model = DecoderOnlyTransformer(
    vocab_size=vocab_size,
    d_model=d_model,
    num_heads=num_heads,
    d_ff=d_ff,
    num_layers=num_layers,
    max_seq_len=max_seq_len
)

input_ids = torch.randint(0, vocab_size, (batch, seq_len))

mask = torch.triu(
    torch.ones(seq_len, seq_len, dtype=torch.bool),
    diagonal=1
)

logits = model(input_ids, mask=mask) # logits: (batch, seq_len, vocab_size)
```

```
next_token_id = top_k_sampling(logits, k=50, temperature=0.8) # next_token_id: (batch,)
```

pytorch 相关基础:

- `nn.Embedding(num_embeddings, embedding_dim)`, 其中 `num_embeddings` 表示嵌入 token 的数量, `embedding_dim` 表示每个 token 映射到多少维度的向量。传入 Embedding 的张量可以是任意维度, 但必须保证张量内每一个数值必须在 0 到 `num_embeddings-1`。
- `torch.Tensor.max(dim)` 返回一个元组: `(values, indices)`, 里面保存了每个仅 `dim` 不同的每一组的最大值及其下标。如果存在多个最大值, 返回的下标是第一次出现最大值的位置。
- 一般 `F.softmax` 本身内部已经做了数值稳定处理, 但如果要显式地展示“防止溢出”技巧, 可以手动先减最大值:

```
def softmax(x, dim=-1):  
    x = x - x.max(dim=dim, keepdim=True).values  
    exp_x = torch.exp(x)  
    return exp_x / exp_x.sum(dim=dim, keepdim=True)
```

- `torch.multinomial()` 会在传入的参数中按数值比例采样, 返回下标。要求输入只能是 1D 或 2D 的非负张量, `multinomial()` 会自动按权重比例采样 (不一定求和要为1)。对于二维输入, 默认对行采样, 采样数量由 `num_samples` 决定。
- `torch.gather(input, dim, index)` 表示沿着 `dim` 维度, 根据 `index` 里的下标, 从 `input` 里取元素。注意 `input` 和 `index` 的维度数必须一样, 输出 `shape` 和 `index.shape` 一样。

3.6 Transformer with KV Cache

带 KV Cache 的 Transformer, 需要使用 2.2 小节中的 `MultiHeadAttention`:

```
import torch  
import torch.nn as nn  
# 自定义模块部分  
from attention import MultiHeadAttention  
from norm import RMSNorm  
from ffn import SwiGLU  
  
class TransformerBlock(nn.Module):  
    def __init__(self, d_model, num_heads, d_ff):  
        super().__init__()  
  
        self.attn = MultiHeadAttention(d_model, num_heads)  
        self.ffn = SwiGLU(d_model, d_ff)  
  
        self.norm1 = RMSNorm(d_model)  
        self.norm2 = RMSNorm(d_model)  
  
    def forward(self, x, mask=None, past_kv=None):  
        """  
        x: (batch, seq_len, d_model)  
        mask: (seq_len, seq_len), causal mask / padding mask  
        """  
        # Pre-LN + MHA + Residual Connection
```

```

    attn_out, new_kv = self.attn(self.norm1(x), mask=mask, past_kv=past_kv)
    x = x + attn_out
    # Pre-LN + FFN + Residual Connection
    x = x + self.ffn(self.norm2(x))
    return x, new_kv

class DecoderOnlyTransformer(nn.Module):
    def __init__(self, vocab_size, d_model, num_heads, d_ff, num_layers, max_seq_len):
        super().__init__()

        self.token_emb = nn.Embedding(vocab_size, d_model) # 词嵌入
        self.pos_emb = nn.Embedding(max_seq_len, d_model) # 可学习的位置编码

        self.layers = nn.ModuleList(
            [TransformerBlock(d_model, num_heads, d_ff) for _ in range(num_layers)]
        ) # 注意力层

        self.norm = RMSNorm(d_model) # 归一化层
        self.lm_head = nn.Linear(d_model, vocab_size, bias=False) # 输出头

    def forward(self, input_ids, mask=None, past_kv=None):
        """
        input_ids: (batch, seq_len), 带KV Cache的情形下一般seq_len=1
        mask: (seq_len, seq_len)
        past_kv: None或list, 长度为num_layers
        past_kv[i]: 第i层的(past_k, past_v)
        past_k/past_v: (batch, num_heads, past_len, head_dim)
        """
        # 获取input_ids相关信息
        seq_len = input_ids.shape[-1]
        device = input_ids.device
        past_len = past_kv[0][0].size(-2) if past_kv is not None else 0

        # 词嵌入并添加可学习位置编码, 注意位置编码应是第past_len到past_len+seq_len之间的位置
        pos = torch.arange(past_len, past_len + seq_len, device=device) # pos: (seq_len,)
        pos = pos.unsqueeze(0) # pos: (1, seq_len), broadcast到batch维度
        x = self.token_emb(input_ids) + self.pos_emb(pos) # x: (batch, seq_len, d_model)

        # 注意力层处理, 处理过程中始终保持x: (batch, seq_len, d_model)
        new_past_kv = []
        for i, layer in enumerate(self.layers):
            layer_past_kv = None if past_kv is None else past_kv[i]
            x, layer_new_kv = layer(x, mask=mask, past_kv=layer_past_kv)
            new_past_kv.append(layer_new_kv)

        x = self.norm(x) # 归一化层
        logits = self.lm_head(x) # 输出头, logits: (batch, seq_len, vocab_size)

        return logits, new_past_kv

```

外部调用:

```
batch = 2
```

```

vocab_size = 10000
d_model = 512
num_heads = 8
d_ff = 2048
num_layers = 6
max_seq_len = 1024

model = DecoderOnlyTransformer(
    vocab_size=vocab_size,
    d_model=d_model,
    num_heads=num_heads,
    d_ff=d_ff,
    num_layers=num_layers,
    max_seq_len=max_seq_len
)

past_kv = None

# 假设逐 token 生成 5 步
for step in range(5):
    input_ids = torch.randint(0, vocab_size, (batch, 1))
    # input_ids: (batch, 1)

    logits, past_kv = model(
        input_ids,
        mask=None,
        past_kv=past_kv
    )

```

4. 损失函数

4.1 SFT (Cross Entropy)

有监督微调 SFT 损失是 next-token prediction 的交叉熵损失：

$$\mathcal{L}_{\text{SFT}} = -\frac{1}{N} \sum_{t \in \text{response tokens}} \log p_{\theta}(y_t \mid x, y_{<t})$$

其中：

- x : prompt
- y_t : 第 t 个目标 token
- $y_{<t}$: 前面已经生成的目标 token

直接使用 torch 中的 `cross_entropy`：

```

import torch.nn.functional as F

# logits: (batch, seq_len, vocab_size)
# labels: (batch, seq_len)

# shift_logits: (batch, seq_len-1, vocab_size), 去掉最后一个token (结束符后面没有token了)
shift_logits = logits[:, :-1, :].contiguous()
# shift_labels: (batch, seq_len-1), 去掉第一个token (起始不需要预测)
shift_labels = labels[:, 1:].contiguous()

loss = F.cross_entropy(shift_logits.view(-1, vocab_size), shift_labels.view(-1),
                       ignore_index=-100)

```

自己实现 `cross_entropy` (一般不强制要求 `ignore_index` 参数) :

```

import torch

def cross_entropy(logits, labels):
    """
    logits: (batch * (seq_len-1), vocab_size)
    labels: (batch * (seq_len-1),)
    """
    # 防止softmax溢出, logits: (batch * (seq_len-1), vocab_size)
    logits = logits - torch.max(logits, dim=-1, keepdim=True).values
    # log_softmax: (batch * (seq_len-1), vocab_size)
    exp_logits = torch.exp(logits)
    probs = exp_logits / exp_logits.sum(dim=-1, keepdim=True)
    log_probs = torch.log(probs)
    # 取对应标签id对应位置的负对数
    n = labels.size(0)
    loss = -log_probs[torch.arange(n, device=labels.device), labels]
    return loss.mean()

# logits: (batch, seq_len, vocab_size)
# labels: (batch, seq_len)

# shift_logits: (batch, seq_len-1, vocab_size), 去掉最后一个token (结束符后面没有token了)
shift_logits = logits[:, :-1, :].contiguous()
# shift_labels: (batch, seq_len-1), 去掉第一个token (起始不需要预测)
shift_labels = labels[:, 1:].contiguous()

loss = cross_entropy(shift_logits.view(-1, vocab_size), shift_labels.view(-1))

```

pytorch 基础:

- `torch.nn.functional.cross_entropy()` 中 `ignore_index=-100` 表示 `label = -100` 的 token 不需要计算 loss, 例如 padding 的地方或者 prompt 就可以设置其 `label = -100`。

4.2 DPO

Direct Preference Optimization, 直接偏好优化。

(1) DPO 数据格式

$$(x, y_w, y_l)$$

- x : prompt
- y_w : chosen, 偏好答案
- y_l : rejected, 不偏好答案

(2) DPO 损失函数 (单样本)

$$\mathcal{L}_{\text{DPO}} = -\log \sigma \left(\beta \left[\log \frac{\pi_{\theta}(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \log \frac{\pi_{\theta}(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right] \right)$$

其中:

- π_{θ} : 当前训练模型
- π_{ref} : 冻结的参考模型, 通常是 SFT 模型
- β : 控制偏好优化强度
- σ : sigmoid

(3) 从隐式 reward 看损失函数

先定义:

$$r_{\theta}(x, y) = \beta [\log \pi_{\theta}(y|x) - \log \pi_{\text{ref}}(y|x)]$$

它可以看成 DPO 的隐式 reward。那么损失就是:

$$\mathcal{L}_{\text{DPO}} = -\log \sigma (r_{\theta}(x, y_w) - r_{\theta}(x, y_l))$$

即 **chosen** 相对于参考模型被提高的水平 - **rejected** 相对于参考模型被提高的水平。因此目标就是提高:

$$r_{\theta}(x, y_w) > r_{\theta}(x, y_l)$$

的概率, 即让 chosen 的相对偏好分数高于 rejected。

```
import torch.nn.functional as F

def dpo_loss(
    policy_chosen_logps,
    policy_rejected_logps,
    ref_chosen_logps,
    ref_rejected_logps,
    beta=0.1
):
    """
    logps: (batch,) 每个值代表每条整段 response 的对数概率。
    """
```

```

# 计算policy和ref各自的chosen-rejected提升概率
policy_logratios = policy_chosen_logps - policy_rejected_logps
ref_logratios = ref_chosen_logps - ref_rejected_logps

# 计算loss
logits = beta * (policy_logratios - ref_logratios)
loss = -F.logsigmoid(logits)
return loss.mean()

```

外部调用:

```

import torch.nn.functional as F

def get_sequence_logps(logits, labels, response_mask):
    """
    logits: (batch, seq_len, vocab_size)
    labels: (batch, seq_len)
    response_mask: (batch, seq_len), 1 表示 response token 参与计算
    """
    shift_logits = logits[:, :-1, :] # shift_logits: (batch, seq_len-1, vocab_size)
    shift_labels = labels[:, 1:] # shift_labels: (batch, seq_len-1)
    shift_mask = response_mask[:, 1:] # shift_mask: (batch, seq_len-1)

    # 转换成对数softmax概率, log_probs: (batch, seq_len-1, vocab_size)
    log_probs = F.log_softmax(shift_logits, dim=-1)

    # 取出response token的对数概率, token_logps: (batch, seq_len-1, vocab_size)
    token_logps = log_probs.gather(
        dim=-1,
        index=shift_labels.unsqueeze(-1) # (batch, seq_len-1, 1), 在vocab维度上gather
    ).squeeze(-1)

    # 过滤non-response token
    sequence_logps = (token_logps * shift_mask).sum(dim=-1)

    return sequence_logps

# 获得policy上chosen和rejected的对数概率得分 (每个response token的对数概率求和)
policy_chosen_logps = get_sequence_logps(policy_chosen_logits, chosen_labels, chosen_mask)
policy_rejected_logps = get_sequence_logps(policy_rejected_logits, rejected_labels,
rejected_mask)

# 获得ref上chosen和rejected的对数概率得分 (每个response token的对数概率求和)
with torch.no_grad():
    ref_chosen_logps = get_sequence_logps(ref_chosen_logits, chosen_labels, chosen_mask)
    ref_rejected_logps = get_sequence_logps(ref_rejected_logits, rejected_labels,
rejected_mask)

# 计算DPO损失
loss = dpo_loss(
    policy_chosen_logps,
    policy_rejected_logps,
    ref_chosen_logps,

```



```
ref_rejected_logps,  
beta=0.1  
)
```

4.3 PPO

Proximal Policy Optimization, 近端策略优化。

(1) 基础的策略梯度

策略模型：

$$\pi_{\theta}(a|s)$$

表示在状态 s 下选择动作 a 的概率。希望高优势的动作概率变大：

$$\mathcal{L}^{PG}(\theta) = \mathbb{E}_t [\log \pi_{\theta}(a_t|s_t) A_t]$$

- $A_t > 0$: 这个动作比平均水平好，概率应提高
- $A_t < 0$: 这个动作差，概率应降低

问题是：一次梯度更新可能把策略改太多，训练不稳定。

(2) PPO 损失函数

$$\text{loss} = \underbrace{-\min(r_t A_t, \text{clip}(r_t, 1-\epsilon, 1+\epsilon) A_t)}_{\text{policy loss}} + \lambda_v \underbrace{(V_{\phi}(s_t) - R_t)^2}_{\text{value loss}} + \lambda_{kl} \underbrace{\text{KL}(\pi_{\theta} \parallel \pi_{\text{ref}})}_{\text{kl loss}}$$

其中：

- r_t : 新旧策略概率比，表示新策略相比于旧策略把动作 a_t 提高/降低的比例。

$$r_t(\theta) = \frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$$

- A_t : 优势函数, $A_t = Q_t(s_t, a_t) - V(s_t)$, 表示当前动作 a_t 比平均预期好多少。
- $V_{\phi}(s_t)$: value model 输出的价值，表示 critic 对状态 s_t 的未来总回报估计。

```
import torch  
  
def mask_mean(loss, mask=None):  
    if mask is not None:  
        return (loss * mask).sum() / mask.sum()  
    else:  
        return loss.mean()  
  
def ppo_loss(  
    new_logprobs,  
    old_logprobs,  
    advantages,  
    values,  
    returns,  
    mask=None,
```

```

ref_logprobs=None,
clip_eps=0.2,
value_coef=0.5,
kl_coef=0.1
):
    """
    new_logprobs: (batch, seq_len) 每时间步下新策略模型输出的对数概率
    old_logprobs: (batch, seq_len) 每时间步下旧策略模型输出的对数概率
    advantages: (batch, seq_len) 每时间步下优势估计
    values: (batch, seq_len) 每时间步下价值模型输出的预测价值
    returns: (batch, seq_len) 每时间步下的GAE回报
    mask: (batch, seq_len), 1 表示有效 token
    ref_logprobs: (batch, seq_len), 每时间步下参考模型输出的对数概率
    返回的是整个batch一起算出的loss
    """
    # 1.policy_loss
    ratio = torch.exp(new_logprobs - old_logprobs) # 新旧策略概率比r_t, 去对数处理

    unclipped = ratio * advantages
    clipped = torch.clamp(ratio, min=1.0-clip_eps, max=1.0+clip_eps) * advantages
    policy_loss = -torch.min(unclipped, clipped)
    policy_loss = mask_mean(policy_loss, mask)

    # 2.value_loss
    value_loss = (values - returns) ** 2
    value_loss = mask_mean(value_loss, mask)

    # 3.kl_loss
    if ref_logprobs is not None:
        log_ratio = ref_logprobs - new_logprobs
        kl_loss = torch.exp(log_ratio) - 1.0 - log_ratio
        kl_loss = mask_mean(kl_loss, mask)
    else:
        kl_loss = 0.0

    # 4.total loss
    loss = policy_loss + value_coef * value_loss + kl_coef * kl_loss

    return loss

```

(3) 广义优势估计

`advantages` 由广义优势估计 GAE 得到，核心思想就是用折扣 TD error 估计优势：

$$\delta_t = R_t + \gamma V(s_{t+1}) - V(s_t)$$

而 $R_t + \gamma V(s_{t+1})$ 正是对 Q_t 的一步 bootstrap 估计。

这里 `rewards` 是每一步 token 的奖励，`returns` 是从该步起的 GAE 回报。

由于 `advantages` 定义为 $A_t = Q_t(s_t, a_t) - V(s_t)$ ，其中 Q_t 是折扣回报 U_t 的期望（在这里还会包含 GAE 平滑，不是一般的折扣回报），因此移项可知：`returns` 可由 `advantages + values` 近似。

GAE 的实现如下：

```

import torch

def advantage_estimate(
    rewards,
    values,
    dones,
    gamma=0.99,
    lam=0.95
):
    """
    rewards/values: (batch, seq_len)
    dones: (batch, seq_len), 1表示该轨迹结束, 后续不再算values, 0表示未结束
    gamma是折扣因子, lam是GAE平滑系数
    """

    # 获取rewards相关信息并初始化
    batch, seq_len = rewards.shape
    device = rewards.device
    advantages = torch.zeros_like(rewards) # advantages: (batch, seq_len)
    last_gae_lam = torch.zeros(batch, device=device) # last_gae_lam: (batch,)

    #
    for t in reversed(range(seq_len)):
        # 计算下一时刻起的values: V(s_{t+1})
        if t == seq_len - 1:
            next_values = torch.zeros(batch, device=device)
        else:
            next_values = values[:, t+1]
        next_non_terminal = 1.0 - dones[:, t] # 是否达到该轨迹末端

        # TD ERROR: delta_t = r_t + gamma * V(s_{t+1}) - V(s_t)
        delta = rewards[:, t] + gamma * next_values * next_non_terminal - values[:, t]

        # 计算GAE
        last_gae_lam = delta + gamma * lam * next_non_terminal * last_gae_lam

        advantages[:, t] = last_gae_lam

    # 返回优势和折扣回报, advantages: (batch, seq_len), returns: (batch, seq_len)
    returns = advantages + values
    return advantages, returns

```

(4) KL 散度的近似

GRPO 损失中的 KL 散度:

$$D_{KL}(\pi_{\theta} \parallel \pi_{\text{ref}}) = \sum_a \pi_{\theta}(a|s) \log \frac{\pi_{\theta}(a|s)}{\pi_{\text{ref}}(a|s)}$$

即:

$$D_{KL}(\pi_{\theta} \parallel \pi_{\text{ref}}) = \mathbb{E}_{a \sim \pi_{\theta}} \left[\log \frac{\pi_{\theta}(a|s)}{\pi_{\text{ref}}(a|s)} \right]$$

由于

$$\text{log_ratio} = \log \frac{\pi_{\text{ref}}}{\pi_{\theta}}$$

所以:

$$D_{KL}(\pi_{\theta} \parallel \pi_{\text{ref}}) = \mathbb{E}_{a \sim \pi_{\theta}}[-\text{log_ratio}]$$

但期望无法直接计算, 因此考虑用蒙特卡洛近似, 而近似值 `log_ratio` 不能保证非负性, 因此需要构造一个新的 KL 散度项来代表 KL 损失。

由于:

$$\mathbb{E}_{a \sim \pi_{\theta}} \left[\frac{\pi_{\text{ref}}(a|s)}{\pi_{\theta}(a|s)} \right] = \sum_a \pi_{\theta}(a|s) \frac{\pi_{\text{ref}}(a|s)}{\pi_{\theta}(a|s)} = \sum_a \pi_{\text{ref}}(a|s) = 1$$

所以:

$$\mathbb{E}_{a \sim \pi_{\theta}} [\exp(\text{log_ratio}) - 1] = 0$$

因此在 KL 的估计量里加上它, 期望不变:

$$\mathbb{E}[-\text{log_ratio}] = \mathbb{E}[\exp(\text{log_ratio}) - 1 - \text{log_ratio}]$$

所以:

$$D_{KL}(\pi_{\theta} \parallel \pi_{\text{ref}}) = \mathbb{E}_{a \sim \pi_{\theta}} [\exp(\text{log_ratio}) - 1 - \text{log_ratio}]$$

就用 `torch.exp(log_ratio) - 1.0 - log_ratio` 来估计 KL 散度。

4.4 GRPO

Group Relative Policy Optimization, 组相对策略优化。

(1) 优势估计

PPO 通常需要如下一系列模型, 比较占用显存:

- policy model
- reference model
- reward model
- value model / critic

GRPO 希望节省 value model 的训练时间和显存占用。

对同一个 prompt x , 一次采样 G 个回答:

$$\{y_1, y_2, \dots, y_G\}$$

分别得到奖励:

$$\{r_1, r_2, \dots, r_G\}$$

然后在这一组内部做标准化:

$$A_i = \frac{r_i - \text{mean}(r_1, \dots, r_G)}{\text{std}(r_1, \dots, r_G)}$$

这个 A_i 就作为第 i 个回答的优势。

(2) GRPO 损失

$$\text{loss} = \underbrace{-\min(r_t A_t, \text{clip}(r_t, 1-\epsilon, 1+\epsilon) A_t)}_{\text{policy loss}} + \lambda_{kl} \underbrace{\text{KL}(\pi_\theta \parallel \pi_{\text{ref}})}_{\text{kl loss}}$$

其中 A_t 由组内相对优势直接得到, 不需要额外的 critic。

```
import torch

def mask_mean(loss, mask=None):
    if mask is not None:
        return (loss * mask).sum() / mask.sum()
    else:
        return loss.mean()

def grpo_loss(
    new_logprobs,
    old_logprobs,
    rewards,
    group_size,
    mask=None,
    ref_logprobs=None,
    clip_eps=0.2,
    kl_coef=0.1,
    eps=1e-8
):
    """
    new_logprobs: (batch, seq_len) 每时间步下新策略模型输出的对数概率
    old_logprobs: (batch, seq_len) 每时间步下旧策略模型输出的对数概率
    rewards: (batch,) 和PPO不同, PPO中returns是逐token的, GRPO中rewards对整个sequence计算
    group_size: GRPO中一组的大小
    mask: (batch, seq_len)
    ref_logprobs: (batch, seq_len) 每时间步下参考模型输出的对数概率
    clip_eps是clip中的eps, eps是计算组间相对优势时标准差用到的eps
    返回的是整个batch一起算出的loss
    """
    # 获取组数
    batch = new_logprobs.shape[0]
    assert batch % group_size == 0
    num_group = batch // group_size

    # 1. 计算相对优势
    group_rewards = rewards.view(num_group, group_size) # group_rewards: (num_group, group_size)
    group_mean = group_rewards.mean(dim=-1, keepdim=True) # group_mean: (num_group, 1)
```

```

    group_std = group_rewards.std(dim=-1, keepdim=True, unbiased=False) # group_std:
(num_group, 1)
    advantages = (group_rewards - group_mean) / (group_std + eps) # advantages: (num_group,
group_size)
    advantages = advantages.view(batch, 1) # advantages: (batch, 1)

# 2. policy_loss
ratio = torch.exp(new_logprobs - old_logprobs)
unclipped = ratio * advantages
clipped = torch.clamp(ratio, min=1.0-clip_eps, max=1.0+clip_eps) * advantages
policy_loss = -torch.min(unclipped, clipped)
policy_loss = mask_mean(policy_loss, mask)

# 3. kl_loss
if ref_logprobs is not None:
    log_ratio = ref_logprobs - new_logprobs
    kl_loss = torch.exp(log_ratio) - 1.0 - log_ratio
    kl_loss = mask_mean(kl_loss, mask)
else:
    kl_loss = 0.0

# 4. total loss
loss = policy_loss + kl_coef * kl_loss
return loss

```

5. 解码策略

5.1 Greedy Decoding

贪心搜索是确定性解码，只选取概率最高的 token 输出。

```

import torch

def greedy_decode(logits):
    """
    logits: (batch, vocab_size)
    return: (batch, 1) 返回next_token
    """
    next_token = torch.argmax(logits, dim=-1, keepdim=True)
    return next_token

```

外部调用（后续 temperature sampling, top-k/p 等调用均类似）：

```

logits = model(input_ids) # logits: (batch, seq_len, vocab_size)

# 只取最后一个token预测
next_token_logits = logits[:, -1, :] # next_token_logits: (batch, vocab_size)

next_token = greedy_decode(next_token_logits) # next_token: (batch, 1)

```

自回归循环生成的版本：

```
import torch

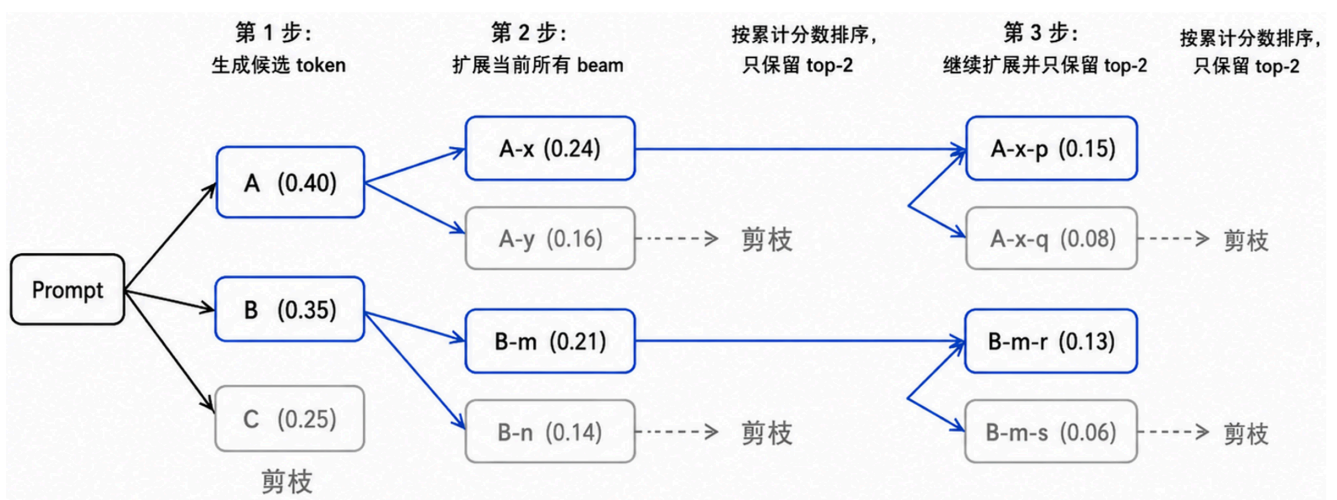
def greedy_search(model, input_ids, max_new_token, eos_token_id=None):
    """
    model: decoder-only LM
    input_ids: (batch, prompt_len) 用户输入的prompt
    max_new_token: 最多生成的新token数量
    eos_token_id: 结束符token id
    return: (batch, seq_len)
    """

    generated = input_ids
    for _ in range(max_new_token):
        # 不用KV Cache的话每步都输入完整序列
        logits = model(generated) # logits: (batch, seq_len, vocab_size)
        # 只取最后一个token预测
        next_token_logits = logits[:, -1, :] # next_token_logits: (batch, vocab_size)
        # 贪心解码
        next_token = torch.argmax(next_token_logits, dim=-1, keepdim=True) # next_token:
        (batch, 1)
        # 在序列维度合并到已生成序列
        generated = torch.cat([generated, next_token], dim=1)
        if eos_token_id is not None and (next_token == eos_token_id).all():
            break
    return generated
```

这里短样本生成 `<eos>` 以后还会继续生成浪费计算，一般实际工程中会再做处理（如维护 `finished` 记录已经生成结束的样本），但在面试手撕时可以简化。

5.2 Beam Search

集束搜索，每次保留多条累计分数最高的生成路径，最终返回分数最高的路径。以下为 `beam_size=2` 的示例。



```
import torch
import torch.nn.functional as F

def beam_search(model, input_ids, max_new_tokens, beam_size=4, eos_token_id=None):
    """
```

```

model: decoder-only LM
input_ids: (batch, prompt_len) 用户输入的prompt
max_new_token: 最多生成的新token数量
beam_size: 每步保留的候选路径数
eos_token_id: 结束符token id
return: (batch, seq_len) 得分最高的序列
"""

batch, prompt_len = input_ids.shape
# 每个样本复制beam_size份
input_ids = input_ids.unsqueeze(1).repeat(1, beam_size, 1) # input_ids: (batch,
beam_size, prompt_len)
input_ids = input_ids.view(batch * beam_size, prompt_len) # input_ids: (batch *
beam_size, prompt_len)
# 初始化beam分数, beam_scores: (batch, beam_size)
beam_scores = torch.zeros(batch, beam_size, device=input_ids.device)
beam_scores[:, 1:] = -1e9 # 初始序列都是prompt, 第一步生成一次即可, 其他分数设为负无穷

for _ in range(max_new_tokens):
    # step1: 生成对数概率
    logits = model(input_ids) # logits: (batch * beam_size, seq_len, vocab_size)
    next_token_logits = logits[:, -1, :] # next_token_logits: (batch * beam_size,
vocab_size)
    logprobs = F.log_softmax(next_token_logits, dim=-1) # logprobs: (batch * beam_size,
vocab_size) 经softmax归一化以后的对数概率
    vocab_size = logits.size(-1)
    logprobs = logprobs.view(batch, beam_size, vocab_size) # logprobs: (batch,
beam_size, vocab_size)

    # step2: 计算各条路径得分
    scores = logprobs + beam_scores.unsqueeze(-1) # scores: (batch, beam_size,
vocab_size)
    scores = scores.view(batch, beam_size * vocab_size) # scores: (batch, beam_size *
vocab_size)

    # step3: 取出得分最高的beam_size条路径信息, 并追溯beam_id, token_id
    # next_scores/ids/beam_ids/tokens: (batch, beam_size)
    next_scores, next_ids = torch.topk(scores, beam_size, dim=-1)
    next_beam_ids = next_ids // vocab_size # 追溯来自哪条beam
    next_tokens = next_ids % vocab_size # 追溯token id

    # step4: 取出得分最高的beam_size条上一轮序列
    input_ids = input_ids.view(batch, beam_size, -1) # input_ids: (batch, beam_size,
seq_len)
    # 为了能从input_ids中gather出next_beam_ids的对应beam, 要保证它们维度相同
    gather_ids = next_beam_ids.unsqueeze(-1).expand(batch, beam_size,
input_ids.size(-1))
    input_ids = torch.gather(input_ids, dim=1, index=gather_ids) # input_ids: (batch,
beam_size, seq_len)

    # step5: 拼接新token并更新beam_scores, 重新展平input_ids作为下一轮模型输入
    input_ids = torch.cat([input_ids, next_tokens.unsqueeze(-1)], dim=-1) # input_ids:
(batch, beam_size, seq_len+1)
    beam_scores = next_scores

```



```

        input_ids = input_ids.view(batch * beam_size, -1) # input_ids: (batch * beam_size,
seq_len)

        if eos_token_id is not None and (next_tokens == eos_token_id).all():
            break

        # 每条样本取分数最高的beam
        input_ids = input_ids.view(batch, beam_size, -1)
        best_seq = input_ids[:, 0, :] # topk会降序排列，取第一个即分数高
        return best_seq

```

pytorch 相关基础:

- `x.repeat(1, beam_size, 1)` 即 `x` 的三个维度分别复制1次、`beam_size` 次、1次。`repeat()` 会将对应序列整体重复生成，例如 `repeat(2)` 将 `[1,2,3]` 重复2次变为 `[1,2,3,1,2,3]`；`repeat_interleave(2, dim=0)` 则会将第 `dim` 维逐元素重复2次，例如 `[1,2,3]` 重复2次变为 `[1,1,2,2,3,3]`。
- `x.expand()` 用于创建广播视图，效果上和 `x.repeat()` 类似，但并不会实际真的复制一份数据，因此更省显存。

5.3 Temperature Sampling

温度采样。高温分布下概率更分散，低温分布下概率更集中。

```

import torch
import torch.nn.functional as F

def temperature_sampling(logits, temperature=1.0):
    """
    logits: (batch, vocab_size)
    return: (batch, 1)
    """
    if temperature <= 0: # temperature=0即贪心搜索
        return torch.argmax(logits, dim=-1, keepdim=True)

    logits = logits / temperature # logits: (batch, vocab_size)
    probs = F.softmax(logits, dim=-1) # probs: (batch, vocab_size)
    next_token = torch.multinomial(probs, num_samples=1) # next_token: (batch, 1)
    return next_token

```

5.4 Top-k Sampling

Top-k 采样。每次选取概率最高的 `k` 个 token 进行采样输出。

```

import torch
import torch.nn.functional as F

def top_k_sampling(logits, k=50, temperature=1.0):
    """
    logits: (batch, vocab_size)
    return: (batch, 1)
    """

```

```

if temperature <= 0: # temperature=0即贪心搜索
    return torch.argmax(logits, dim=-1, keepdim=True)

# 温度+top-k采样, topk_logits/idx: (batch, k)
logits = logits / temperature
topk_logits, topk_idx = torch.topk(logits, k, dim=-1)

# 只对top-k进行softmax归一化, 并取下标映射回原始id
probs = F.softmax(topk_logits, dim=-1) # probs: (batch, k)
sampled_idx = torch.multinomial(probs, num_samples=1) # sampled_idx: (batch, 1)
next_token = torch.gather(topk_idx, dim=-1, index=sampled_idx) # next_token: (batch, 1)
return next_token

```

5.5 Top-p Sampling

Top-p 采样。每次从概率高到低选取累计概率至少覆盖 p 的 token 进行采样输出。

```

import torch
import torch.nn.functional as F

def top_p_sampling(logits, p=0.9, temperature=1.0):
    """
    logits: (batch, vocab_size)
    """
    if temperature <= 0: # temperature=0即贪心搜索
        return torch.argmax(logits, dim=-1, keepdim=True)

    # 温度采样和降序排序, sorted_logits/idx: (batch, vocab_size)
    logits = logits / temperature
    sorted_logits, sorted_idx = torch.sort(logits, dim=-1, descending=True)

    # 计算累计概率, sorted_probs / cum_probs: (batch, vocab_size)
    sorted_probs = F.softmax(sorted_logits, dim=-1)
    cum_probs = torch.cumsum(sorted_probs, dim=-1)

    # 大于p的位置使用掩码-inf, 并至少保留第一个超过p的token, 使累计概率覆盖p
    sorted_mask = cum_probs > p
    sorted_mask[:, 1:] = sorted_mask[:, :-1].clone() # 全体右移1位
    sorted_mask[:, 0] = False # 第一个位置不掩码
    sorted_logits = sorted_logits.masked_fill(sorted_mask, float('-inf'))

    # 只对top-p进行softmax归一化, 并进行采样, 映射回原始下标
    probs = F.softmax(sorted_logits, dim=-1) # probs: (batch, vocab_size)
    sampled_idx = torch.multinomial(probs, num_samples=1) # sampled_idx: (batch, 1)
    next_token = torch.gather(sorted_idx, dim=-1, index=sampled_idx) # next_token: (batch,
1)
    return next_token

```

6. 其他模块

6.1 LoRA

LoRA (Low-Rank Adaptation) 微调对原模型的权重矩阵 W 添加两个可训练的低秩矩阵 A 和 B , 形成增量:

$$W' = W + \Delta W = W + BA$$

其中 $A \in \mathbb{R}^{r \times d}$, $B \in \mathbb{R}^{d \times r}$, $r \ll d$ 。

通过低秩矩阵学习特定任务, 并保持原模型权重冻结, 从而减少显存占用。LoRA 原文建议的初始化是 A 随机, B 为 0, 这样让 LoRA 分支初始输出为 0, 从而一开始完全不扰动原始预训练模型。

```
class LoRALinear(nn.Module):
    def __init__(self, base_linear, r):
        """
        base_linear: nn.Linear(d_in, d_out)
        """
        super().__init__()

        self.base = base_linear
        for p in self.base.parameters():
            p.requires_grad = False

        in_dim = base_linear.in_features
        out_dim = base_linear.out_features

        self.A = nn.Linear(in_dim, r, bias=False)
        self.B = nn.Linear(r, out_dim, bias=False)

        nn.init.zeros_(self.B.weight)  # A初始随机化, B初始为0

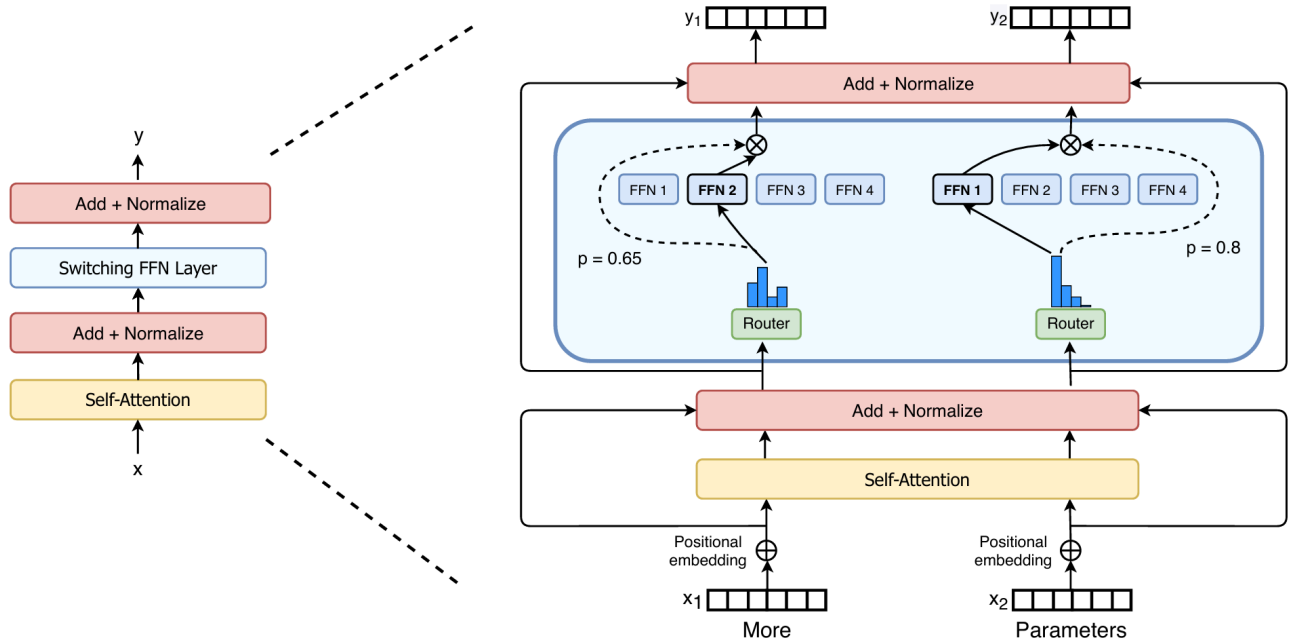
    def forward(self, x):
        return self.base(x) + self.B(self.A(x))
```

6.2 MoE

混合专家模型。

(1) MoE 架构

本质上 MoE 是在 Transformer 的 FFN 层上做稀疏化扩展。先由 Router 为每个 token 计算各专家得分, 再选取得分最高的 Top-k 专家处理, 并按路由权重聚合输出。



```
import torch
import torch.nn as nn
import torch.nn.functional as F

class Expert(nn.Module):
    def __init__(self, d_model, d_ff):
        super().__init__()
        self.net = nn.Sequential(
            nn.Linear(d_model, d_ff),
            nn.GELU(),
            nn.Linear(d_ff, d_model)
        )

    def forward(self, x):
        """
        x: (..., d_model)
        return: (..., d_model)
        """
        return self.net(x)

class MoE(nn.Module):
    def __init__(self, d_model, d_ff, num_experts, top_k=2):
        super().__init__()
        self.num_experts = num_experts
        self.top_k = top_k

        self.router = nn.Linear(d_model, num_experts, bias=False)
        self.experts = nn.ModuleList([Expert(d_model, d_ff) for _ in range(num_experts)])

    def forward(self, x):
        """
        x: (batch, seq_len, d_model)
        return: 线性层输出和负载均衡损失, out: (batch, seq_len, d_model), aux_loss
        """
```

```

# x按token展平
batch, seq_len, d_model = x.shape
x_flat = x.view(batch * seq_len, d_model) # x_flat: (batch * seq_len, d_model)

# 计算路由到各个专家的分數, router_logits: (batch * seq_len, num_experts)
router_logits = self.router(x_flat)

# 每个token选取top-k专家并重新归一化, topk_logits/idx/probs: (batch * seq_len, top_k)
topk_logits, topk_idx = torch.topk(router_logits, self.top_k, dim=-1)
topk_probs = F.softmax(topk_logits, dim=-1)

# 计算负载均衡损失, moe_aux_loss之后定义
aux_loss = moe_aux_loss(
    router_logits=router_logits,
    topk_idx=topk_idx,
    num_experts=self.num_experts
)

# 初始化输出, out_flat: (batch * seq_len, d_model)
out_flat = torch.zeros_like(x_flat)

# 对每个专家, 处理各自的token
for expert_id, expert in enumerate(self.experts):
    mask = topk_idx == expert_id # 哪些token选中当前的专家
    if not mask.any():
        continue
    # 对于编号为token_idx的token, 当前专家是它第which_k个top-k专家
    # token_idx和which_k是长度为selected_len的向量, selected_len为选中当前专家的token数
    token_idx, which_k = torch.where(mask)
    # 专家处理, input: (selected_len, d_model), output: (selected_len, d_model)
    expert_input = x_flat[token_idx]
    expert_output = expert(expert_input)
    # 对于每个token, 将权重回传, weight: (selected_len, 1)
    weight = topk_probs[token_idx, which_k].unsqueeze(-1)
    out_flat.index_add_(dim=0, index=token_idx, source=expert_output * weight)

# 将输出形状调整回去, out: (batch, seq_len, d_model)
out = out_flat.view(batch, seq_len, d_model)
return out, aux_loss

```

pytorch 相关基础:

- `nn.Sequential` 用于定义线性流水线模块, 使用时会按顺序执行每一层。 `Expert` 中的 `self.net` 等价于:

```

self.fc1 = nn.Linear(d_model, d_ff)
self.act = nn.GELU()
self.fc2 = nn.Linear(d_ff, d_model)

def forward(self, x):
    x = self.fc1(x)
    x = self.act(x)
    x = self.fc2(x)
    return x

```

- `nn.GELU` 即 Gaussian Error Linear Unit，高斯误差线性单元。 $GELU(x) = x \Phi(x)$ ，其中 $\Phi(x)$ 是标准正态分布的累积分布函数。 x 越大，其权重 $\Phi(x)$ 越大，越会被保留，反之则会被抑制。
- `torch.where(mask)` 返回 True 位置的下标 tuple，`torch.where(mask, x, y)` 在 mask 为 True 的地方返回 x，在 mask 为 False 的地方返回 y。
- `Tensor.index_add_(dim, index, source)` 用于把 source 里的内容，按照 index 指定的位置，累加到张量的第 dim 维上，相当于 pytorch 版本的 `array[idx] += value`。

(2) Auxiliary Loss

负载均衡损失，防止路由器把大量 token 都分给少数几个 expert，导致 expert 失衡甚至出现专家塌缩。负载均衡损失最后加到训练总损失中。

$$L_{\text{aux}} = \alpha N \sum_{i=1}^N f_i P_i$$

其中：

- N ：expert 数量
- f_i ：实际被分配到 expert i 的 token 比例
- P_i ：router 给 expert i 的平均概率
- α ：aux loss 的权重

当所有 expert 完全均衡时， $f_i = \frac{1}{N}$ ， $P_i = \frac{1}{N}$ ，此时 aux loss 取最小值：

$$L_{\text{aux}} = \alpha N \cdot N \cdot \frac{1}{N^2} = \alpha$$

当所有 token 都路由到同一个 expert，且 router 概率也全部压到它上面时， $f_1 = 1$ ， $P_1 = 1$ ，其他 expert 都是 0，此时 aux loss 取最大值：

$$L_{\text{aux}} = \alpha N \cdot 1 = \alpha N$$

```

import torch
import torch.nn.functional as F

def moe_aux_loss(router_logits, topk_idx, num_experts):
    """
    router_logits: (num_tokens, num_experts)
    topk_idx: (num_tokens, top_k)
    """

```

```

# 1.Router概率pi
router_probs = F.softmax(router_logits, dim=-1) # router_probs: (num_tokens,
num_experts)
pi = router_probs.mean(dim=0) # pi: (num_experts,)

# 2.token分配比例fi
expert_mask = F.one_hot(topk_idx, num_classes=num_experts).float() # expert_mask:
(num_tokens, top_k, num_experts)
fi = expert_mask.mean(dim=(0, 1)) # fi: (num_experts,)

aux_loss = num_experts * torch.sum(pi * fi)
return aux_loss

```

pytorch 相关基础:

- `torch.nn.functional.one_hot()` 会把 `(num_tokens, top_k)` 形状的 `topk_idx` 的值变成一个个 one-hot 向量, 从而形状变为 `(num_tokens, top_k, num_experts)`。 `one_hot()` 要求给定张量的值必须在 `0-num_classes-1` 之间, 否则会报错。另外 `one_hot()` 返回的是整数, 为了后续能够计算均值, 要转为 `float()`。